

A probabilistic analysis of the maximal covering location problem

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Abstract

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Under a variety of different random models of the maximal covering location problem, we show that the relative error of a randomly generated solution converges to zero in expectation as problem size grows. We prove similar results for the relative error between the optimal integer and fractional solutions to the problem. Suppose that randomly generated instances of this problem are used to test heuristics. One consequence of our results is that we should expect the mean relative error of a heuristic to be better than that of randomly generated solutions, if the heuristic is to be considered useful.

1. Introduction

The maximal covering location problem (MCP) is a mathematical problem used in deciding the location of emergency facilities such as fire stations and oil spill response centers [2, 4, 11]. We are given a collection of m demand centers (say, potential oil spill regions) and a set of n potential facility locations. Each possible facility location is said to cover a group of demand centers if it can service, or meet the demands of, these centers. Associated with each demand center is a positive

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number or weight w_i , representing the importance of covering that demand center. The object is to select K ($\leq n$) of the potential facility locations so as to maximize the sum of weights of demand centers covered.

To formulate MCP as a mathematical program let:

$$x_j = \begin{cases} 1, & \text{if facility site } j \text{ is chosen, } j = 1, \dots, n, \\ 0, & \text{otherwise,} \end{cases}$$

$$y_i = \begin{cases} 1, & \text{if demand center } i \text{ is covered, } i = 1, \dots, m, \\ 0, & \text{otherwise,} \end{cases}$$

$$a_{ij} = \begin{cases} 1, & \text{if facility } j \text{ covers demand center } i, i = 1, \dots, m, j = 1, \dots, n, \\ 0, & \text{otherwise.} \end{cases}$$

Then MCP takes the form:

$$\begin{aligned} Z = \max \quad & \sum_{i=1}^m w_i y_i, & (\text{MCP}) \\ \text{s.t.} \quad & \sum_{j=1}^n a_{ij} x_j \geq y_i, \quad i = 1, \dots, m, \\ & \sum_{j=1}^n x_j = K, \\ & x_j = 0, 1, \quad j = 1, \dots, n, \\ & y_i = 0, 1, \quad i = 1, \dots, m. \end{aligned}$$

Clearly, a polynomial algorithm for MCP with arbitrary value of K would imply a polynomial algorithm for the set covering problem which is well known, [6], to be NP-hard. Hence MCP is NP-hard. In this paper we will analyze, from both a worst-case and a probabilistic viewpoint, the behavior of some heuristics for MCP. Specifically, in Section 2 we describe a simple greedy heuristic for MCP that produces a solution no smaller than $\max\{K/n, 1 - (1 - 1/K)^K\}$ times the optimal. In Sections 3, 4 and 5 we prove, under three different random models of MCP, that the relative error of a randomly generated solution converges to zero in expectation as the size of the problem increases. Suppose that randomly generated instances of MCP are used for the testing of heuristics. Then an important consequence of this paper is that the results of such an empirical test will only be taken to show good performance (with respect to mean relative error) if those results provide evidence of performance superior to a randomly generated solution. We will also show that the relative difference between optimal integer and fractional solutions converges to zero in expectation. This last result provides some theoretical confirmation for a

conclusion reached by empirical observation, [10]. For results of a similar flavor for related problems the interested reader is referred to [1, 6, 7, 12, 13].

2. The greedy heuristic

The first heuristic we study is a simple “greedy” one. Before describing it we introduce some notation. For any $S \subseteq \{1, 2, \dots, m\}$ let $w(S) = \sum_{i \in S} w_i$. For all $j = 1, \dots, n$ let $S_j = \{i: a_{ij} = 1\}$.

GREEDY.

Step 1. Set $r=0$, $x_j=0$, $j=1, \dots, n$.

Step 2. Find the index $q \in \{1, \dots, n\}$ that maximizes $w(S_j)$.

Step 3. Set $x_q = 1$ and $r = r + 1$. If $r = K$, stop, feasible solution found.

Step 4. For all $j \in \{1, \dots, n\}$, set $S_j = S_j \setminus S_q$ and return to Step 2.

Let Z_G denote the value of the solution $x_1, x_2, \dots, x_n$ returned by GREEDY, i.e., the total weight of demand centers covered.

Theorem 1. $Z_G/Z \geq \max\{K/n, 1 - (1 - 1/K)^K\}$.

Proof. Let $f(C) = \sum_{i=1}^m w_i \max_{j \in C} a_{ij}$ for any set C of facility sites. Then, it is easy to see that $f(\emptyset) = 0$ and that f is submodular, i.e., $f(A) + f(B) \geq f(A \cup B) + f(A \cap B)$ for all $A, B \subseteq \{1, \dots, n\}$. Hence,

$$\begin{aligned} Z &= \max_C f(C), \\ \text{s.t. } & |C| = K, \\ & C \subseteq \{1, \dots, n\}. \end{aligned}$$

Problem MCP is thus the problem of maximizing a submodular function and GREEDY is the standard myopic heuristic for this problem. So, invoking a result of [5, 9] we obtain:

$$Z_G/Z \geq 1 - \left(1 - \frac{1}{K}\right)^K.$$

Let the indices selected in Step 2 of GREEDY be numbered $1, 2, 3, \dots, K$. Let S_j^r denote the set S_j at the r th iteration of Step 2. Observe that the sets $S_1^1, S_2^2, \dots, S_n^n$ (if we allowed GREEDY to continue despite the fact that $r > K$) form a partition of $\{1, \dots, m\}$. Hence, by Step 2 of the heuristic:

$$Z_G = \sum_{r=1}^K w(S_r^r) \geq \frac{K}{n} \sum_{r=1}^n w(S_r^r) \geq \frac{K}{n} \sum_{i=1}^m w_i.$$

However, $Z \leq \sum_{i=1}^m w_i$. Thus, $Z_G/Z \geq K/n$. Clearly, Z_G/Z must be at least as large as the larger of its two lower bounds, hence:

$$Z_G/Z \geq \max \left\{ \frac{K}{n}, 1 - \left(1 - \frac{1}{K} \right)^K \right\}. \quad \square$$

3. Random model I

In the first random model of MCP we assume the w_i are independent identically distributed random variables in $[0, 1]$ with finite mean μ and variance σ^2 . We also assume that for $1 \leq i \leq m$, $1 \leq j \leq n$, $a_{ij} = 1$ with probability p , and $a_{ij} = 0$ with probability $1 - p$, independently of the other a_{ij} . This allows for the possibility that the covering matrix $A = (a_{ij})$ contains a row of zero entries. To prevent this, we eliminate any row containing only zero entries, and generate the row anew until it contains at least one nonzero entry. This has the effect of making the entries in a row dependent, but entries in different rows are still independent. We shall show that a randomly generated solution to MCP under these assumptions is likely to be optimal. Results of this kind have been obtained in [7, 12] for the set covering problem, under essentially the same assumptions.

Let R be any set of K columns of the A matrix. Without loss of generality we can take R to be columns $\{1, 2, \dots, K\}$. Let $Z_r = \sum_{i=1}^m w_i \max_{j \in R} a_{ij}$. Clearly, Z_r is the value of a randomly generated solution.

Theorem 2. $E(Z_r)/E(Z) \geq 1 - (1 - p)^K$.

Proof.

$$\begin{aligned} E(Z_r) &= E \left(\sum_{i=1}^m w_i \max_{j \in R} a_{ij} \right) \\ &= \sum_{i=1}^m \mu E \left(\max_{j \in R} a_{ij} \right) \\ &\geq m\mu(1 - (1 - p)^K) \end{aligned}$$

because there is at least one nonzero entry in row i . Now, $E(Z) \leq m\mu$ and so the result follows. $\square$

Note that Theorem 2 does not allow us to conclude that $Z_r/Z \rightarrow 1$ in expectation as $n, K \rightarrow \infty$. However, the next result does so.

Theorem 3. $E(Z_r/Z) \geq 1 - (1 - p)^K$.

Proof. Let $Q_i = \max_{j \in R} a_{ij}$, $i = 1, \dots, m$, and $L = (\sum_{i=1}^m w_i Q_i)(\sum_{i=1}^m w_i)^{-1}$. Clearly $Z_r/Z \geq L$. Let $t_1, \dots, t_m$ be m numbers in $[0, 1]$. We will compute the expectation of L conditional on w_i falling in a small interval above t_i , for $i = 1, \dots, m$. Then,

$$\begin{aligned}
E(L \mid t_i \leq w_i \leq t_i + \Delta t_i \quad \forall i) &\leq E\left(\sum_{i=1}^m \frac{(t_i + \Delta t_i)Q_i}{\sum_{i=1}^m t_i}\right) \\
&= \left[\sum_{i=1}^m (t_i + \Delta t_i)E(Q_i)\right] \left[\sum_{i=1}^m t_i\right]^{-1}.
\end{aligned}$$

Similarly,

$$E(L \mid t_i \leq w_i \leq t_i + \Delta t_i \quad \forall i) \geq \left[\sum_{i=1}^m t_i E(Q_i)\right] \left[\sum_{i=1}^m t_i + \Delta t_i\right]^{-1}.$$

Now,

$$\begin{aligned}
E(L) &= \lim_{\Delta t_i \rightarrow 0} \int_{t_1=0}^1 \cdots \int_{t_m=0}^1 E(L \mid t_i \leq w_i \leq t_i + \Delta t_i \quad \forall i) \\
&\quad \times \Pr(t_i \leq w_i \leq t_i + \Delta t_i \quad \forall i) dt_1 dt_2 \cdots dt_m \\
&\leq \int_{t_1=0}^1 \cdots \int_{t_m=0}^1 E(Q_i) \Pr(w_i \leq t_i \quad \forall i) dt_1 \cdots dt_m \\
&= E(Q_i) \int_0^1 \cdots \int_0^1 \prod_{i=1}^m \Pr(w_i \leq t_i) dt_i \\
&= E(Q_i).
\end{aligned}$$

Similarly, $E(L) \geq E(Q_i) \Rightarrow E(L) = E(Q_i) \geq 1 - (1-p)^K$. $\square$

In order to study the asymptotic behavior of Z and Z_r , we need to specify how m and K behave as n grows. We shall assume that both m and K grow with n . We do not restrict in any way the rate at which they grow.

Theorem 4. $Z_r/m, Z/m \rightarrow \mu$ almost surely as $n \rightarrow \infty$.

Proof. $Z_r/m = (1/m) \sum_{i=1}^m w_i Q_i$. Now, $w_i Q_i$ can be treated as a single random variable and it has finite mean. So, by the Law of Large Numbers, $Z_r/m - E(w_i Q_i) \rightarrow 0$ almost surely as $m \rightarrow \infty$. But $E(w_i Q_i) \rightarrow \mu$ as $K \rightarrow \infty$. Since by assumption m and $K \rightarrow \infty$ as $n \rightarrow \infty$, it follows that $Z_r/m \rightarrow \mu$ almost surely as $n \rightarrow \infty$.

Now $Z_r/m \leq Z/m \leq (1/m) \sum_{i=1}^m w_i \rightarrow \mu$ almost surely as $n \rightarrow \infty$. Hence, $Z/m \rightarrow \mu$ almost surely as $n \rightarrow \infty$. $\square$

Corollary. $Z_r/Z \rightarrow 1$ almost surely as $n \rightarrow \infty$.

Thus, the random selection heuristic is asymptotically optimal. It has been observed [10] that the LP solution to MCP is frequently very close to the optimal integer solution. Our next result attempts to make this observation precise. Let Z_{LP} denote the optimal objective function value to MCP when the integrality constraints are dropped.

Theorem 5. For m fixed and $K < n$, $m\mu - E(Z_{LP}) \rightarrow 0$ as $K, n \rightarrow \infty$.

Proof. Clearly $E(Z_{LP}) \leq m\mu$. Consider the solution $x_j = K/n$ for all j . This is clearly feasible for the LP relaxation of MCP. Hence

$$Z_{LP} \geq \sum_{i=1}^m w_i \min \left\{ 1, \frac{K}{n} \sum_{i=1}^m a_{ij} \right\}.$$

Let $v = \lceil n/K \rceil$. A straightforward calculation shows that

$$E(Z_{LP}) \geq m\mu \left\{ \sum_{j=v}^n \binom{n}{j} p^j (1-p)^{n-j} + \frac{K}{n} \sum_{j=1}^{v-1} \binom{n}{j} p^j (1-p)^{n-j} \right\}.$$

Hence:

$$\begin{aligned} m\mu - E(Z_{LP}) &\leq m\mu \left\{ \sum_{j=1}^n \binom{n}{j} p^j (1-p)^{n-j} - \sum_{j=v}^n \binom{n}{j} p^j (1-p)^{n-j} \right. \\ &\quad \left. - \frac{K}{n} \sum_{j=1}^{v-1} \binom{n}{j} p^j (1-p)^{n-j} \right\} \\ &= m\mu \left\{ \left(1 - \frac{K}{n} \right) \sum_{j=1}^{v-1} \binom{n}{j} p^j (1-p)^{n-j} \right\}. \end{aligned}$$

For K sufficiently large, $1/K < p$, so,

$$m\mu - E(Z_{LP}) \leq m\mu \left(1 - \frac{K}{n} \right) \sum_{j=1}^{\lfloor np \rfloor} \binom{n}{j} p^j (1-p)^{n-j} \rightarrow 0$$

as $n \rightarrow \infty$ by the Chernoff bounds [3]. $\square$

It is now an easy matter to deduce that $E(Z)/E(Z_{LP}) \rightarrow 1$ as m, K and $n \rightarrow \infty$.

4. Random model II

It has been pointed out, [10], that instances of MCP where the column sums of the A matrix are close to being equal are much more difficult to solve computationally than when there is great variability in the column sums. The reason for this is that the gap between Z_{LP} and Z tends to be larger than usual, so increasing the amount of time branch and bound or dual ascent procedures take to solve MCP. We introduce random model II as a model of instances of MCP where the column sums of A are close to being equal.

As before we assume the w_i are independently and identically distributed in $[0, 1]$ with finite mean μ . In this second model, the number of ones in each column of A will be treated as independent draws from a Poisson distribution with mean λ , where $\lambda = \lfloor cm \rfloor$ and $c < 1$. Once the number of nonzero entries for each column has been specified, those entries will be distributed with equal probability among the m rows of that column. Note that this procedure is not guaranteed to produce a matrix without a row of zeros.

We will use the same notation as in Section 3. The analysis for this model differs in the calculation of $\Pr(\max_{j \in R} a_{ij} = 1)$.

Lemma 6.

$$\Pr\left(\max_{j \in R} a_{ij} = 1\right) \geq 1 - \left[1 - \frac{\lambda}{m} \sum_{r=1}^{m-1} \frac{e^{-\lambda} \lambda^{r-1}}{(r-1)!}\right]^K.$$

Proof. We will compute $\Pr(\max_{j \in R} a_{ij} = 0)$. Because the nonzero entries in each column are no longer independent,

$$\begin{aligned} \Pr\left(\max_{j \in R} a_{ij} = 0\right) &\leq \prod_{j \in R} \Pr(a_{ij} = 0) \\ &= [\Pr(a_{i1} = 0)]^K \\ &= \left[\sum_{r=0}^{m-1} \Pr\left(a_{ij} = 0 \mid \sum_{k=1}^m a_{i1} = r\right) \frac{e^{-\lambda} \lambda^r}{r!} \right]^K \\ &= \left[\sum_{r=0}^{m-1} \frac{(m-r)}{m} \frac{e^{-\lambda} \lambda^r}{r!} \right]^K \\ &= \left[1 - \frac{\lambda}{m} \sum_{r=1}^{m-1} \frac{e^{-\lambda} \lambda^{r-1}}{(r-1)!} \right]^K. \end{aligned}$$

Using the fact that $\Pr(\max_{j \in R} a_{ij} = 1) = 1 - \Pr(\max_{j \in R} a_{ij} = 0)$, we get the required result. $\square$

Let $\alpha_m = \sum_{r=1}^{m-1} e^{-\lambda} \lambda^{r-1} / (r-1)!$. It is easy to see that $\lim_{m \rightarrow \infty} \alpha_m = 1$. Now using the same arguments as in Theorems 2 and 3 we obtain the following:

$$\frac{E(Z_r)}{E(Z)} \geq 1 - \left(1 - \frac{\lambda \alpha_m}{m}\right)^K, \quad (1)$$

$$E\left(\frac{Z_r}{Z}\right) \geq 1 - \left(1 - \frac{\lambda \alpha_m}{m}\right)^K. \quad (2)$$

Notice that the right-hand side quantities of (1) and (2) go to 1 as $n, K, m \rightarrow \infty$.

5. Random model III

The third random model we consider is a geometrical one. Here we place n points independently and uniformly in the unit square. These points represent both demand centers and potential facility locations. Label these points $1, \dots, n$. A point will “cover” another point if the two are at most d units in distance apart. In effect $a_{ij} = 1$ if i and j are at most d units apart for all $i, j = 1, \dots, n$, and $a_{ij} = 0$ otherwise. We make the same assumption about the distribution of the w_i as in Sections 3 and 4.

Two observations are in order before proceeding. If d is fixed or grows with n

then, for K sufficiently large, MCP becomes trivial to solve. Alternatively, if d declines too rapidly with n , say $d^2 = n^{-1-\varepsilon}$ then with probability tending to one as $n \rightarrow \infty$, $a_{ij} = 0$ for all $i, j = 1, \dots, n$. We therefore want d^2 to decrease with n (or K) but not as rapidly as $n^{-1-\varepsilon}$.

Theorem 7. $E(Z_r)/E(Z) \rightarrow 1$ for $d = O(1/K)$, $K = O(\sqrt{n})$.

Proof.

$$\begin{aligned} E(Z_r) &= E\left(\sum_{i=1}^n w_i \max_{j \in R} a_{ij}\right) \\ &= E\left(\sum_{i \in R} w_i\right) + E\left(\sum_{i \notin R} w_i \max_{j \in R} a_{ij}\right) \quad \text{as each point } i \in R \text{ covers itself,} \\ &= K\mu + (n-K)\mu E\left(\max_{j \in R} a_{K+1,j}\right). \end{aligned}$$

The probability that a single point whose label is in the set R is not within d units of the point labeled $K+1$ is $\leq 1 - \pi d^2/4$. Hence, the probability that at least one point in R is within d units of $K+1$ is $\geq 1 - (1 - \pi d^2/4)^K$.

Thus, $E(\max_{j \in R} a_{K+1,j}) \geq 1 - (1 - \pi d^2/4)^K$, and so:

$$\begin{aligned} E(Z_r) &\geq K\mu + (n-K)\mu \left[1 - \left(1 - \frac{\pi d^2}{4}\right)^K\right] \\ &= K\mu \left(1 - \frac{\pi d^2}{4}\right)^K + n\mu \left[1 - \left(1 - \frac{\pi d^2}{4}\right)^K\right]. \end{aligned}$$

Hence,

$$\frac{E(Z_r)}{E(Z)} \geq \frac{K}{n} \left(1 - \frac{\pi d^2}{4}\right)^K + 1 - \left(1 - \frac{\pi d^2}{4}\right)^K \rightarrow 1 \quad \text{as } n \rightarrow \infty,$$

hence the result follows. $\square$

Observe that under random model III,

$$\begin{aligned} \Pr\left(\sum_{j=1}^n a_{ij} = r\right) &= \Pr(r \text{ points cover point } i) \\ &= \binom{n-1}{r-1} \Pr(\text{a point is within distance } d \text{ of } i)^{r-1} \\ &\quad \times [1 - \Pr(\text{a point is within distance } d \text{ of } i)]^{n-r} \\ &\geq \binom{n-1}{r-1} \left(\frac{\pi d^2}{4}\right)^{r-1} (1 - \pi d^2)^{n-r}. \end{aligned}$$

Hence, using an argument similar to the one used in Theorem 5, we can prove that $E(Z)/E(Z_{LP}) \rightarrow 1$ as $n \rightarrow \infty$ as well.

Conclusions

We have shown that the relative error of a randomly generated solution to MCP converges to zero in expectation as problem size grows. This result is quite robust, as it holds for at least three different random models of MCP. An interesting question for future investigation is whether these results can be strengthened to almost sure convergence. It would also be interesting to study the behavior of solutions for MCP under random model III for growth rates of K other than $O(\sqrt{n})$.

References

- [1] S. Ahn, C. Cooper, G. Cornuejols and A. Frieze, Probabilistic analysis of a relaxation for the k -median problem, *Math. Oper. Res.* 13 (1988) 1–31.
- [2] S. Belardo, J. Harrauld, W. Wallace and J. Ward, A partial covering approach to siting response resources for major maritime oil spills, *Management Sci.* 30 (1984) 1184–1196.
- [3] H. Chernoff, A measure of asymptotic efficiency for tests of a hypothesis based on the sum of observations, *Ann. Math. Statist.* 23 (1952) 493–507.
- [4] R. Church and C. Reville, The maximal covering location problem, *Papers Regional Sci. Appl.* 32 (1974) 101–118.
- [5] M. Conforti and G. Cornuejols, Submodular functions, matroids and the greedy algorithm: tight worst-case bounds and some generalizations of the Rado–Edmonds theorem, *Discrete Appl. Math.* 7 (1984) 257–274.
- [6] M.R. Garey and D.S. Johnson, *Computers and Intractability: A Guide to the Theory of NP-Completeness* (Freeman, New York, 1979).
- [7] D.S. Hochbaum, Easy solutions for the K -center problem or the dominating set problem on random graphs, in: *Annals of Discrete Mathematics* 25 (North-Holland, Amsterdam, 1985) 189–210.
- [8] A. Jain, Probabilistic analysis of LP relaxation bounds for the Steiner tree problem in networks, *Networks*, to appear.
- [9] G.L. Nemhauser and L. Wolsey, Maximizing submodular set functions: formulations and analysis of algorithms, in: *Studies on Graphs and Discrete Programming* (North-Holland, Amsterdam, 1981) 279–301.
- [10] C. Skiscim, Personal communication, 1986.
- [11] C. Toregas, R. Swain, C. Reville and L. Bergman, The location of emergency facilities, *Oper. Res.* 19 (1971) 1363–1373.
- [12] C. Vercellis, Probabilistic analysis of set covering, *Ann. Oper. Res.* 1 (1984) 255–271.
- [13] R.V. Vohra and D. Foster, A probabilistic analysis of the K -location problem, *Amer. J. Math.*, to appear.